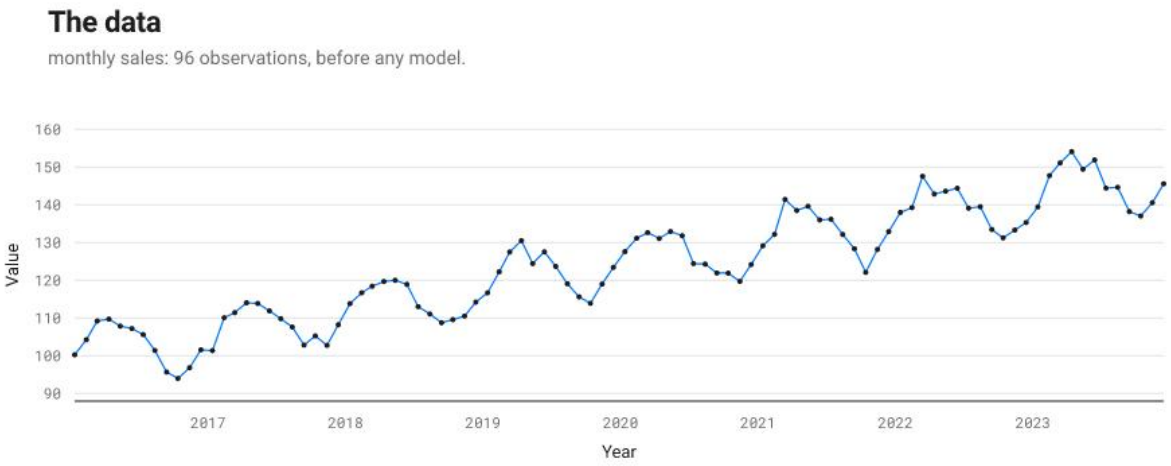
An automatic report for the dataset: monthly sales

Some Math empowered by some AI guided by some researchers

09 November 2026

Produced by **`autostat`***, a Python 3 port of the Automatic Bayesian Covariance Discovery system of Lloyd et al. (AAAI 2014).*



The data

Executive summary

The structure search selected `Lin + WN + Per` (Bayesian information criterion 449.2; negative log likelihood 210.9 on 96 points). It decomposes into 3 additive components, listed by how much each reduces extrapolation error:

1. A linearly increasing function.
2. A periodic function with a period of 1.0 years.
3. Uncorrelated noise.

#	R^2 cumulative (%)	Residual variance explained (%)	CV MAE	MAE reduction (%)
data			124	
1	83.8	83.8	5.16	95.8
2	98.4	90.3	1.58	69.3
3	100.0	100.0	1.58	-0.0

MAE: mean absolute error over ten contiguous held-out blocks; CV: cross-validated.

The formula found

The structure search is a symbolic regression: it searches a grammar of expressions over kernels and keeps the one with the lowest Bayesian information criterion. This is the expression it kept, with every fitted parameter in place; t is in years.

$$y(t) = f_1(t) + f_2(t) + f_3(t)$$

Each component is a Gaussian process with its own covariance:

$$f_1(t) \sim \mathrm{GP}\big(0,\ \ 37.14\,(t-2000)(t'-2000)\big)$$

$$f_2(t) \sim \mathrm{GP}\left(0,\ \ 31.71\,\frac{\exp\Big(0.01596\cos\frac{2\pi(t-t')}{1}\Big) - I_0(0.01596)}{e^{0.01596} - I_0(0.01596)}\right)$$

$$f_3(t) \sim \mathrm{GP}(0,\ \ 3.578\,\delta_{tt'})$$

These components have a closed form, read from their posterior mean:

$$f_1(t) = 99.63 + 6.095\,(t-2016)$$

$$f_2(t) \approx 7.95\cos\left(\frac{2\pi\cdot(t-2016)}{1} - 1.807\right) + 0.3961\cos\left(\frac{2\pi\cdot 2\cdot(t-2016)}{1} - 0.9546\right)$$

A periodic kernel is exactly a sum of cosines, so a periodic component is a Fourier series in its period; it is written up to the harmonic after which the rest weighs less than 0.1

The other components are smooth, quasi-periodic or noise without a short formula: a draw from such a process has no finite closed form; its covariance above is the formula.

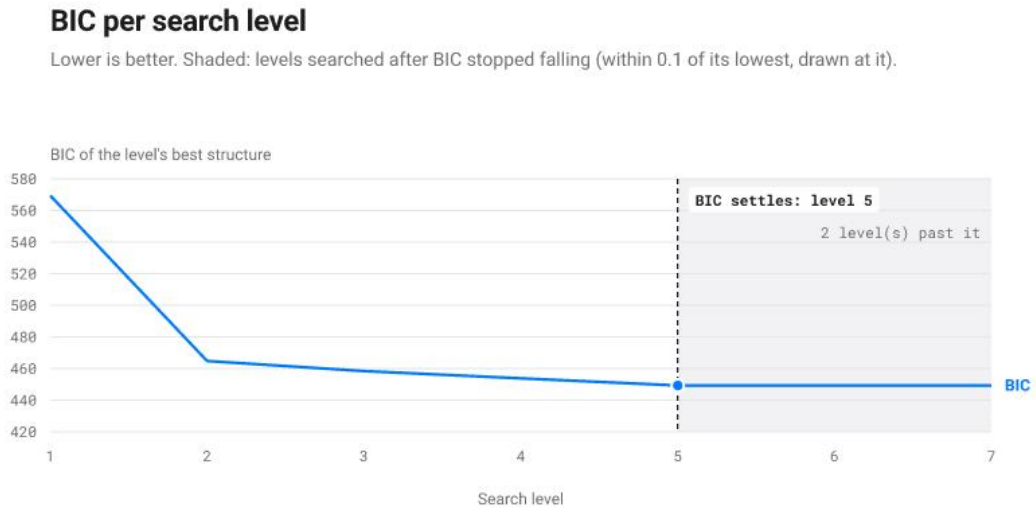
Closed form of each component:

#	component	closed form
1	A linearly increasing function	exact polynomial of degree 1
2	A periodic function with a period of 1.0 years	Fourier series, 2 harmonics, largest gap 0.027 % of the component
3	Uncorrelated noise	none: the covariance is the formula

Search path and stopping criterion

The 2014 configuration searches a fixed number of levels and keeps the lowest BIC; it has no stopping rule. The table follows the winner of each level. A change of structure adds or removes terms; a level that keeps the structure only refits its parameters. BIC differences read on the scale of Kass and Raftery (1995): under 2 barely worth mentioning, 2 to 6 positive, 6 to 10 strong, over 10 very strong.

level	BIC	change	structure	components	closed	params
1	569.4		WN + SE	2	0	3
2	464.8	-104.6	WN + Per x SE + SE	3	0	7
3	458.4	-6.4	WN + Lin x SE + Per x SE	3	0	8
4	453.8	-4.6	Lin + WN + Per x SE	3	1	7
5	449.2	-4.6	Lin + WN + Per	3	2	6
6 (kept)	449.2	-0.0	Lin + WN + Per	3	2	6
7	449.2	+0.0	Lin + WN + Per	3	2	6

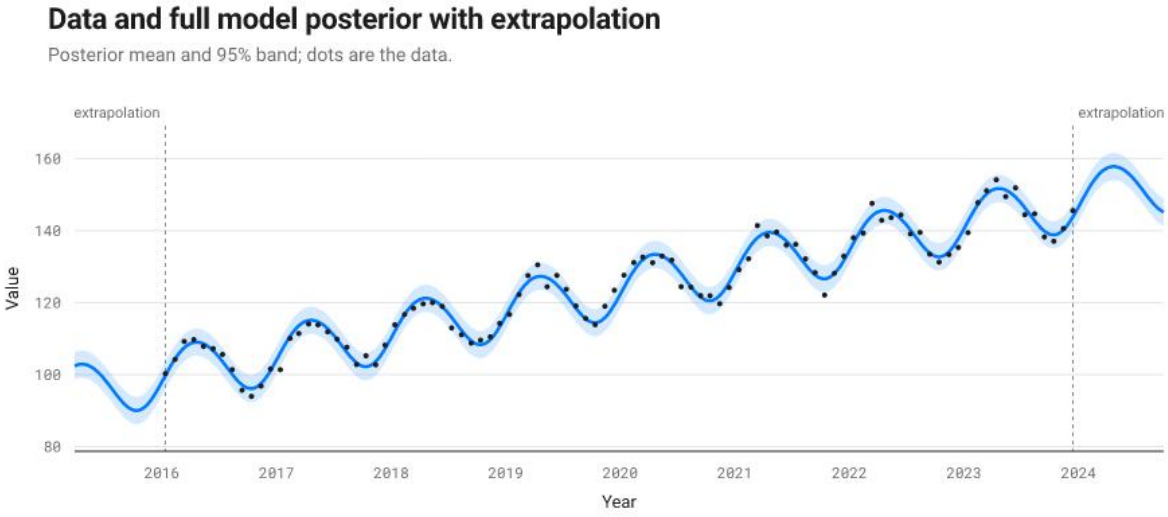


BIC per search level

Posterior weight of each structure met, exp(-BIC/2) normalised under a flat prior, after the Bayesian machine scientist (Guimerà et al., 2020): where the search keeps one structure, the weights say how much the data prefer it to the others. The 2 structure(s) under 0.001 are left out.

structure	lowest BIC	weight
Lin + WN + Per	449.2	0.899
Lin + WN + Per x SE	453.8	0.092
WN + Lin x SE + Per x SE	458.4	0.009

- The search ran 7 levels (stopping after 2 level(s) without a BIC fall above 0.1, at most 20) and keeps the model of level 6, Lin + WN + Per, BIC 449.2.
- That structure was first reached at level 5; the levels after it only refitted its parameters, lowering BIC by 0.0 in all.
- The best different structure, Lin + WN + Per x SE (level 4), is 4.6 above it: positive evidence for the kept one on the Kass and Raftery scale.
- Weighed by exp(-BIC/2) under a flat prior, as the Bayesian machine scientist weighs expressions, the structures met give the kept one a posterior weight of 0.899 and the next, Lin + WN + Per x SE, 0.092: the data favour the kept one without ruling the next out.
- BIC stopped falling at level 5. The 2 level(s) searched past it kept its structure and lowered BIC by 0.00 in all, less than the tolerance of 0.1: going on buys nothing the data pay for.
- A rule stopping after 1 level(s) without a fall above 0.1 would have ended the search at level 6 and kept the same model.
- A rule stopping after 2 level(s) without a fall above 0.1 would have ended the search at level 7 and kept the same model.
- The kept expression has 6 parameters for 96 points: its BIC is a fit term 2 NLL = 421.9 plus a complexity term p ln n = 27.4, each extra parameter costing 4.56; it splits into 3 components, 2 of them with a closed form.



Data and full posterior

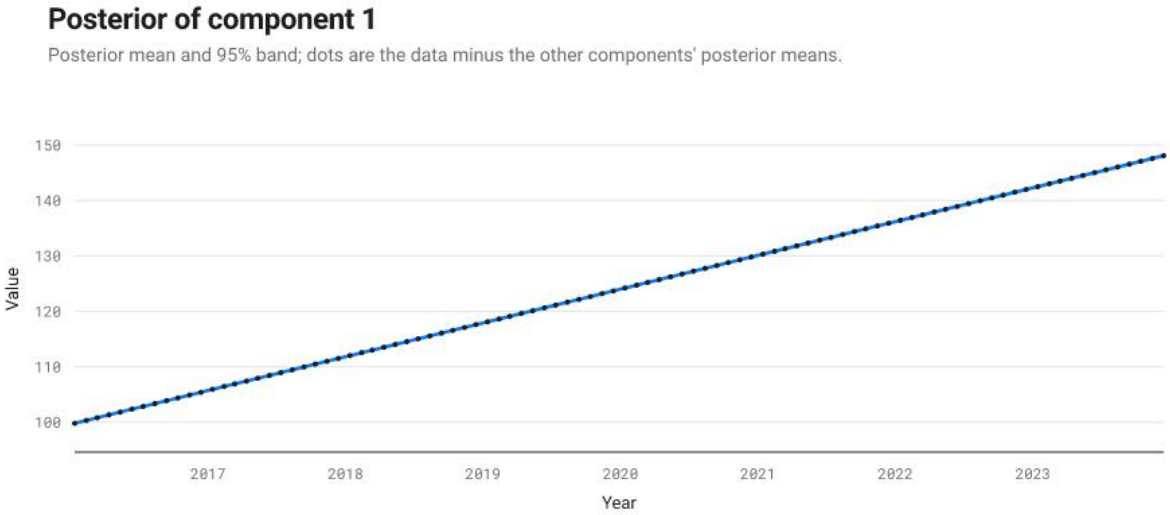
Detailed discussion of the components

Component 1: A linearly increasing function

This component is linearly increasing.

This component explains 83.8% of the data's variance on its own; with the components above it, 83.8%. Adding it changes the cross-validated MAE by -95.8%.

Extrapolation. This component is assumed to continue to increase linearly.



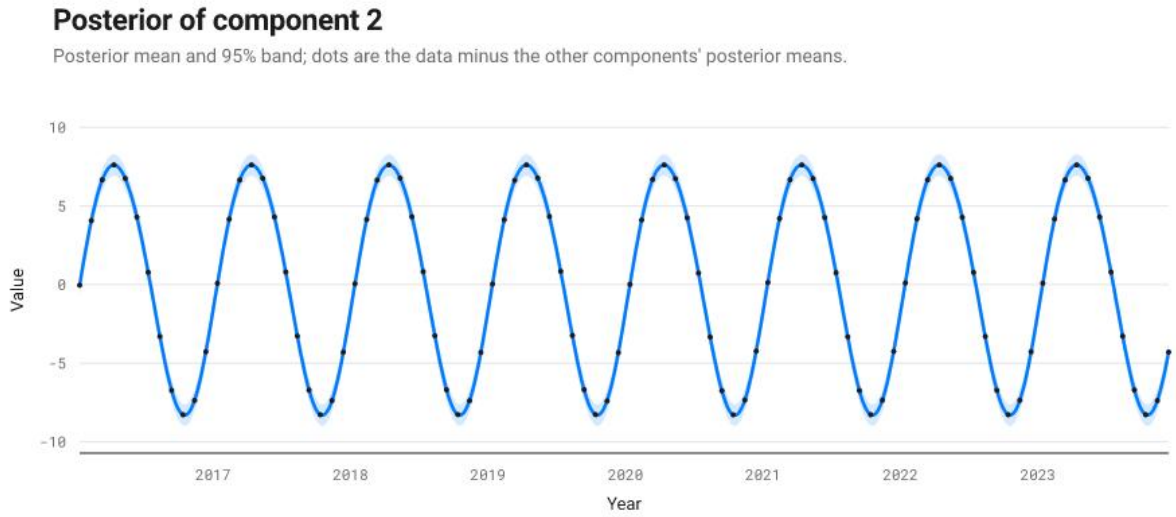
Component 1

Component 2: A periodic function with a period of 1.0 years

This component is periodic with a period of 1.0 years. The shape of this function within each period is very smooth and resembles a sinusoid.

This component explains 7.7% of the data's variance on its own; with the components above it, 98.4%. Adding it changes the cross-validated MAE by -69.3%.

Extrapolation. This component is assumed to continue periodically with a period of 1.0 years.



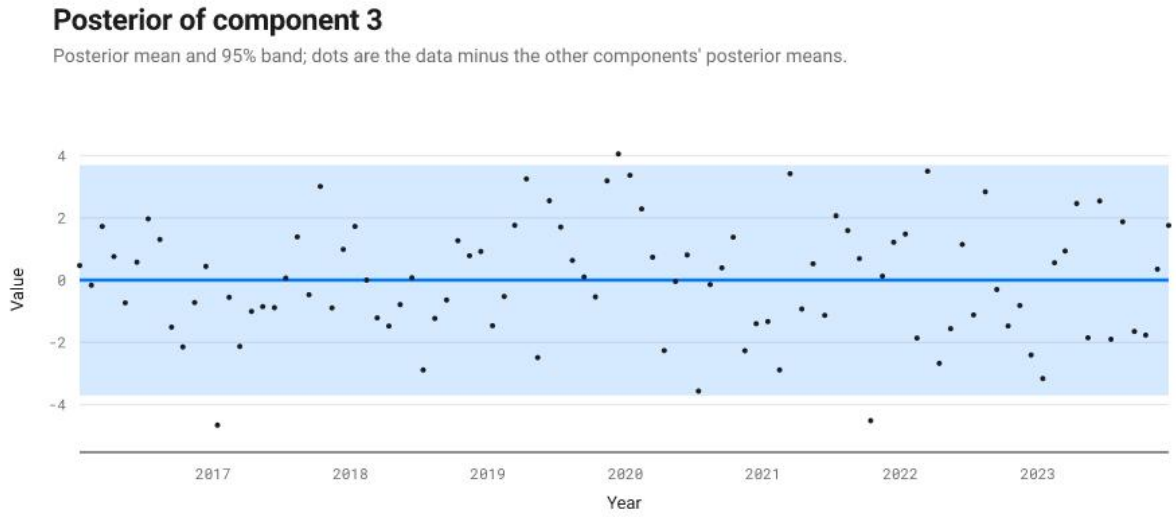
Component 2

Component 3: Uncorrelated noise

This component models uncorrelated noise.

This component explains 1.7% of the data's variance on its own; with the components above it, 100.0%. Adding it changes the cross-validated MAE by 0.0%.

Extrapolation. This component assumes the uncorrelated noise will continue indefinitely.



Component 3

Model checking

Each statistic compares draws of a component's posterior with draws of its prior. A small p-value (bold below 0.05) points at a property of the data that the component's prior does not reproduce.

#	QQ max	QQ min	ACF min	ACF min lag	Pgram max	Pgram peak	MMD
1	0.348	0.348	0.964	0.980	0.668	0.966	**0.000**
2	0.352	0.522	0.850	0.978	0.744	0.954	**0.010**
3	0.508	0.876	0.974	0.974	0.980	0.962	**0.003**

QQ: quantile-quantile gaps. ACF: autocorrelation function. Pgram: periodogram. MMD: maximum mean discrepancy two-sample test.